

# Victor Fournet

## Curriculum Vitæ

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### Personal Information

**Name:**, Victor Fournet  
**Age:**, 28 years old, born October 16, 1997 in Paris  
**Status:**, Single, no children  
**Nationality:**, French

### Professional Experience

Since February 2025 **Postdoctoral Researcher**, Max Planck Institute for Plasma Physics, Garching, Germany  
November 2021 **PhD Student**, CEA-DAM-DIF, France  
– January 2025

### Education

2021–2025 **PhD in Mathematics**, CEA-DAM-DIF, Sorbonne Université  
Supervisors Bruno Després, Christophe Buet  
Subject **Mathematical and Numerical Study of Thick Spray Models**  
Defense January 22, 2025  
Committee Emmanuel Grenier, *Professor*, Beijing Institute of Technology, Reviewer  
Julien Mathiaud, *Professor*, Université de Rennes (IRMAR), Reviewer  
Laurent Boudin, *Associate Professor*, Sorbonne Université (LJLL)  
Frédérique Charles, *Professor*, Université Grenoble Alpes (LJK), Committee Chair  
Bruno Després, *Professor*, Sorbonne Université (LJLL), PhD Supervisor  
Christophe Buet, *Research Director*, CEA-DAM-DIF, PhD Co-supervisor

2020–2021 **Master's Degree in Mathematics**, Sorbonne Université, Paris, France  
"Mathématiques de la Modélisation"

2019–2020 **Master's – Agrégation Preparation**, Sorbonne Université, Paris, France

2018–2019 **First Year of Master's Degree in Mathematics**, Sorbonne Université, Paris, France

2015–2018 **Bachelor's Degree in Mathematics**, Sorbonne Université, Paris

2015–2018 **Bachelor's Degree in Physics**, Sorbonne Université, Paris

### Internships

January 2018 – **L3 Internship supervised by Didier Smets**, Laboratoire Jacques Louis Lions  
April 2018 (LJLL), Sorbonne Université, Paris  
Subject: *Eikonal Equation and application to shape from shading.*

January 2019 – **M1 Internship supervised by Vincent Humilière**, Institut Paris Rive Gauche,  
June 2019 Sorbonne Université, Paris  
Subject: *The h-principle*

April 2021 – **M2 Internship supervised by Christophe Buet and Stéphane Del Pino,**  
September *French Alternative Energies and Atomic Energy Commission (CEA)*  
2021 Subject: *Construction, analysis and implementation of a nodal finite volume scheme for the  $P_N$  model.*

## Technical Skills

Systems **Unix/Linux, Mac OS, Windows**  
Languages **Python, basic knowledge of C++**  
Parallel **Basic knowledge of MPI**  
Computing

## Languages

**English,** *Read, written and spoken*  
**Spanish,** *Basic knowledge*

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## Publications

### Published peer-reviewed journal articles

- [1] B. Després, C. Buet and V. Fournet, *Local-in-time existence of strong solutions to an averaged thick sprays model*. Kinetic and Related Models, 17(4): 606-633, 2024. HAL version.
- [2] B. Després, C. Buet and V. Fournet, *Analog of Linear Landau Damping in a coupled Vlasov-Euler system for thick sprays*. Communications in Mathematical Sciences, 23: 235-258, 2025.
- [3] C. Buet, S. Del Pino and V. Fournet. *Construction, analysis and implementation of two nodal finite volume schemes for the  $P_N$  model for particle transport in 2D*. The SMAI Journal of Computational Mathematics, Volume 11 (2025), pp. 39-84. doi: 10.5802/smai-jcm.119

### Accepted peer-reviewed journal articles

- [4] D. Bian, B. Després, V. Fournet and E. Grenier, *Landau damping in mixed hyperbolic-kinetic systems and thick sprays*. Accepted in Kinetic and Related Models.

### Published peer-reviewed proceedings

- [5] V. Fournet, Investigation on the stability in a thick spray model. ESAIM: PROCEEDINGS AND SURVEYS, 2025, Vol. 79, p. 31-41. doi: 10.1051/proc/202579031.
- [6] Buet, Christophe, Bruno Després, and Victor Fournet. "A Coupled Semi-Lagrangian/Finite." In Rarefied Gas Dynamics: Proceedings of the 33rd International Symposium, p. 511. Springer Nature, 2026. <https://doi.org/10.1007/978-3-032-00094-1>

### Articles in preparation

- [7] M. Campos-Pinto, V. Fournet, E. Franck, and V. Michel-Dansac, *Non-linear control variate in  $\delta f$  particle-in-cell methods using symplectic neural networks*. In preparation.

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## Scientific Presentations

### Contributed talks at conferences

- [1] *Some new results on the numerical simulation of thick sprays*, SMAI 2023, Pointe à Pitre, Guadeloupe, May 2023.
- [2] *Coupling kinetic and fluid equation: Landau damping in fluid particles model*, Conference for Young Researchers in Mathematics and Applications 2023, Gif-sur-Yvette, France, September 2023.
- [3] *Coupling kinetic and fluid equation: Landau damping in fluid particles model*, Numkin 2023, Max Planck Institute for Plasma Physics, Garching, Germany, November 2023.
- [4] *Coupling kinetic and fluid equations: Linear (in)stability in fluid particles interactions*, CANUM 2024, Le Bois-Plage-en-Ré, France, May 2024.
- [5] *Coupling kinetic and fluid equations: Close packed limit in thick sprays*, RGD 2024, Göttingen, Germany, June 2024.
- [6] *Linear Landau damping in thick sprays*, SMAI 2025, Bordeaux, France, June 2025.
- [7] *Non-linear control variate in  $\delta f$  methods using symplectic neural networks*, SIAM Conference on Parallel Processing for Scientific Computing (PP26), Berlin, Germany, March 2026.

### Contributed talks at workshops

- [8] *An averaged model for thick spray*, Workshop: Kinetic and Hyperbolic Equations – Modelling, Analysis and Numerics, Institut de Mathématiques de Toulouse, Toulouse, France, December 2022.
- [9] *Linear Landau damping in thick sprays*, Journées Jeunes EDPiste, Université Paris Dauphine-PSL, Paris, France, January 2026.

### Seminar talks

- [10] *Coupling kinetic and fluid equations in the context of sprays: Landau damping and close-packed limit*, PhD students' working group, Laboratoire Jacques Louis Lions, Paris, France, November 2023.
- [11] *Linear damping in Fluid-Kinetic models for thick sprays*, NMPP Seminar, Max Planck Institute for Plasma Physics, Garching, Germany, June 2024.

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## Teaching Activities

2022 **Analysis Course**, *Tutorial sessions for first-year undergraduates (53h)*, Sorbonne Université, September–December

2018–2020 **Mathematics Tutoring**, *Exercise sessions for second and third-year undergraduates*, Sorbonne Université